Ship Classification in Medium-Resolution SAR Images via Densely Connected Triplet CNNs Integrating Fisher Discrimination Regularized Metric Learning

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***Abstract*— Thanks to the medium-to-high resolution and wide coverage imaging ability, satellite synthetic aperture radar (SAR) systems are momentously developed for intelligent maritime surveillance, especially for ship classification in SAR images. Previous researchers mostly utilized the geometric, radiometric, and structural features combined with the traditional machine-learning (ML) methods to conduct ship classification in high-resolution (HR) SAR images. However, the handcrafted features showed weak representation for the medium-resolution (MR) SAR images, and the normal ML methods are less powerful to deal with the intraclass diversity and interclass similarity in the MR SAR ship images. To address these issues, this article extends the dense convolutional networks to the MR SAR ship classification for better deep features extraction and proposes a multitask learning framework within which the softmax log-loss and the triplet loss are jointly minimized for more effective ship classification in MR SAR images. The triplet loss is computed by imposing a similarity constraint on the deep embeddings via the deep metric learning (DML) scheme such that the deep representations of the same class are pulled much closer to each other and those from different classes are pushed as farther apart as possible. However, the triplet loss in the usual DML utilizes the triplets mined in a training batch independently, which ignores the contextual information. So, we propose to impose a Fisher discrimination regularization term on the deep embeddings to explore the global information of learned embeddings, and hence the designed task-specific networks will be trained to have more robust and better recognition performance. Extensive experiments on the MR SAR ship data set collected from the Sentinel-1 SAR images demonstrate that the proposed method can achieve superior performance for the 3- and 5-class recognition tasks with regard to the comparing CNN models.**

***Index Terms*— Densely connected convolutional networks, Fisher discrimination, medium resolution (MR), metric learning, ship classification, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR), due to its noticeable advantages of working all-day and in various weather conditions, has become a significant device for many remote sensing applications [1]–[5]. Of the most important ones, ship surveillance over wide marine areas using space-borne SAR systems is highly demanded for most of the authorities [4]–[15]. To name a few, national defense intelligence, fisheries monitoring and enforcement, search and rescue support, and shipping traffic are the most applicable fields [16]–[19]. With the advent of more advanced satellite SAR systems, an abundance of valuable data are easy to be acquired, especially those from the littoral areas and the open sea. Thus, high precision and operational ship surveillance algorithms, which are able to comprehensively interpret such massive maritime data, deserve to be developed.

Conventionally, a general-purpose ship surveillance system mainly consists of four stages [4], [11], [17]: land masking, prescreening, discrimination, and classification. The first stage usually uses a shoreline database to acquire accurate land-sea segmentation for better ship detection over the sea area [4], [17], [20]. Prescreening is primarily to obtain the candidate ship detections using mostly the constant false alarm rate algorithms in single-channel or polarimetric SAR images [4], [21]–[23]. In the discrimination stage, some target measurements representing ship characteristics are utilized to discriminate the true ship detections from the artifacts in order to further reduce the false alarms [4], [20], [24], [25]. Classification, where an *in-situ* specific category will be assigned to each ship target [10]–[13], has received less attention due to the absence of available benchmarking data set and the low resolution (LR) of the early SAR systems. With the advent of more medium-to-high resolution SAR systems, ship classification using such data has attracted more attention. Regarding the focus of this article, the medium resolution (MR) SAR ship classification methods will be fully investigated.

Fig. 1 shows several challenging issues in the MR SAR ship classification. First, compared to the high-resolution (HR) SAR images, ship targets in LR and MR SAR images only possess a few pixels and present spot-like appearances. Thus, it is difficult to exactly extract the regular ship representations,

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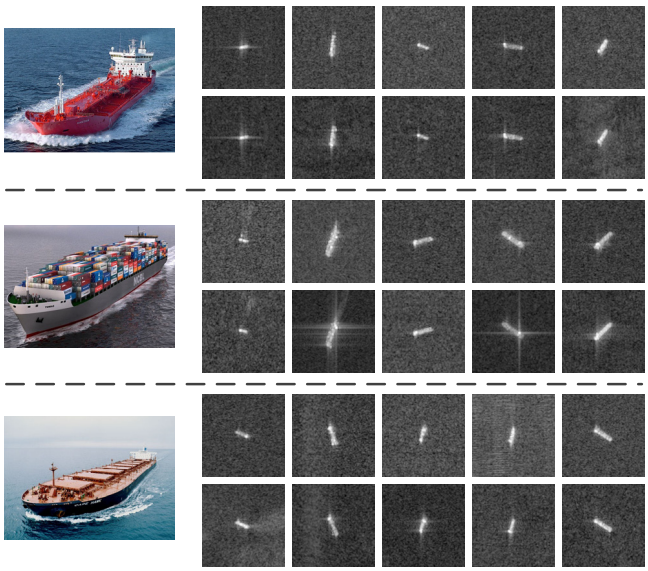


Fig. 1. Optical photographs and the corresponding SAR chips of (Top) tanker, (Middle) container ship, and (Bottom) bulk carrier. The SAR ship chips are extracted from the Sentinel-1 images. For each ship type in one panel, the chips of the first and second rows are in VH and VV polarizations, respectively. One can find that large within-class diversity and between-class similarity exist in this data set which will severely degenerate the recognition performance.

such as the length, width, aspect ratio [19], and mean intensity. Second, owing to the complex imaging conditions and diversified looking angles between the ship targets and the SAR sensors, ships in SAR images appear in various aspects with respect to the azimuth direction and different scattering characteristics. Hence, it is kind of difficult to represent such rich semantic information using the low-level geometric and radiometric features [12], [13], [18], [19], [26]. Due to the impacts of the factors above, large intraclass variation and small interclass separation present. In Fig. 1, different ship samples of each type have various sizes, different orientation angles, and diverse scattering characteristics, and the same sample in different polarization modes also presents distinct scattering properties. Moreover, strong similarities exist among different ship types in terms of the abovementioned properties. Therefore, it is worthwhile to develop more discriminative ship representations in MR SAR images.

To address the challenges, this article resorts to the convolutional neural networks (CNNs) [27]–[30]. Compared with the low-level features [12], [13], [18], [19], [26] and the mid-level image representations [25], the CNNs [27]–[30] can learn more nonlinear hierarchical deep features directly from large scale data and have broad applications in computer vision [31]–[35] as well as the SAR automatic target recognition [36]–[39]. However, there exist two shortcomings to apply the plain CNNs for MR SAR ship classification. First, as shown in [39] and [41]–[43], the CNNs trained with the common softmax log-loss [37], [44] cannot explicitly leverage the learning of discriminative features. Second, the volume of SAR ship data is relatively small, e.g., the pilot and passenger vessels in the OpenSARShip data set [18] only have 14 and 44 samples, respectively. Both of the issues will prevent the plain CNNs from classifying the SAR ship targets effectively. In this article, a densely connected CNN structure based on the advanced dense convolutional networks (DenseNets) [40] is developed for a more effective representation of MR SAR images. The core idea of DenseNet is the dense connection from the shallow layers to all the deeper ones, by which more new features can be explored to represent the ship targets with less parameters required [40].

To further boost the feature discrimination, the task-specific densely connected CNN is combined with the triplet networks [45], [46] to address the intraclass diversity and interclass similarity via the deep metric learning (DML) scheme [47]–[52]. Compared to the normal CNNs, the triplet CNNs comprise three identical networks with each taking input an image of a triplet. One triplet is built by sampling an anchor image, a positive image of the same class to the anchor and a negative image of another class, from the training batch. Then, the sampled triplets are fed into the triplet CNNs to compute the triplet loss [33], by minimizing which the positive and anchor images are brought closer in the learned feature space and the negative ones are pushed far apart from the anchors such that the intraclass compactness and interclass scatter are explicitly encouraged, rendering the subsequent classification much simpler [53]. Moreover, lots of triplets sampled can compensate the low number of samples in some classes [48], [49] for more discriminative features learning. To integrate the merits of the triplet loss into the softmax loss, the multitask learning framework [42], [43] is used to simultaneously optimize these two losses. Furthermore, to consider the holistic correlation of the triplets in each training batch, we propose a new regularization term inspired by the Fisher discriminant [54] and add it to the objective function to make full use of the triplets in training batches for robust generalization. To summarize, our main contributions are as follows.

1) A novel and task-specific backbone network based on the DenseNets is designed to extract powerful deep features, which is then embedded into the triplet network to perform ship classification in MR SAR images.
2) DML is first introduced to improve the intraclass compactness and interclass separation for effective MR SAR ship classification. To mitigate the overfitting of the triplet loss, we develop a new Fisher regularization term to utilize the global information in the mined triplets, which leads to better and more robust performance.
3) Based on the OpenSARShip data set and the Sentinel-1 images, a systemic reprocessing procedure is developed to collect the experimental data. Extensive experiments validate the superior performance of the proposed ship classification method over the comparing methods.

The remainder of this article is organized as follows. Section II briefly reviews the most related works of our method. Section III gives some preliminary knowledge for the proposed method. Section IV describes the proposed ship classification method in detail and implementation details are introduced in Section V. Extensive experiments are provided in Section VI. Finally, Section VII provides the conclusion and outlines future works.

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II. Related Work

This section presents an overview of the ship classification in SAR images and the DML, respectively.

A. Ship Classification in SAR Imagery

Margarit, Mallorqui, and Fabregas [10], Margarit and Mallorqui [11], Margarit, Mallorqui, Lopez-Martinez *et al.* [12], and Margarit and Tabasco [13] had conducted milestone works for ship classification in SAR images, such as initially generating simulated ship samples for ship category classification [10] and proposing to divide the segmented ship signature into the bow, middle, and stern sections to derive a parametric vector for ship representation [13], which instigated a series of later works [55]–[60]. Furthermore, Zhang *et al.* [61] and Wang *et al.* [62] took an analysis of the scattering components of the ship hulls to classify the merchant ships into bulk carriers, container ships, and oil tankers. In brief, the previous ship classification methods applied more handcrafted features, such as the geometric and scattering features [55]–[65], to represent the ship signatures. However, these low-level features are less discriminative and generally validated by the limited HR SAR data.

Recently, some attempts aiming at improving SAR ship classification using the deep learning methods have been implemented [66]–[72]. Specifically, to alleviate the issue of the limited data volume, Ødegaard *et al.* [67] proposed to use some simulated samples for better network training, and Wang *et al.* [71] employed the transfer-learning and fine-tuning techniques to perform ship classification with small data sets. To explore more details of the ship targets, Bentes *et al.* [70] developed a multiple-input resolution CNN model for better deep features extraction. Regarding the recently released OpenSARShip data set [18], different ship types are unbalanced, and the ship chip sizes are different from each other. Li *et al.* [72] exploited the upsampling and the ratio batching techniques to deal with the issues above. However, the inherent details of the ships shall degenerate in the upsampling process. In this article, based on the OpenSARShip ground truths and the original Sentinel-1 database, we propose to recollect the data set to acquire the ship chips with the same size.

B. Deep Metric Learning

DML aims to combine the distance metric learning [73] and the deep networks [27]–[30] for more effective deep embedding learning [74]–[80]. Two commonly used DML paradigms are the triplet networks [46] and the siamese networks [74]. Generally, the triplet network with triplet loss [33] has a better performance, which leads to its broad applications in visual recognition and retrieval tasks [76]–[78]. To improve the representation of deep models, Hoffer and Ailon [46] and Parkhi *et al.* [75] proposed to fuse the classification task and the triplet embedding learning in a consecutive fashion, which may show suboptimal performance. In this article, similar to [42] and [43], we propose an end-to-end multitask learning scheme for MR SAR ship classification by jointly optimizing the Softmax classifier [44] and the triplet loss simultaneously.

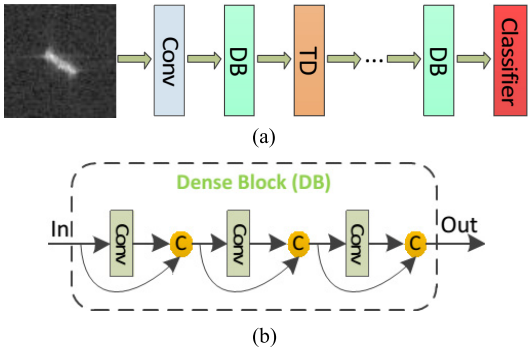


Fig. 2. (a) Architecture of the DenseNet. An input image is first processed through a normal convolution layer (Conv), and the output feature maps are fed into the consecutive DB and TD modules. TD consists of a Conv layer and a pooling layer. (b) Detailed illustration of the DB. The Conv combined with the preceding concatenation operation C incarnate the dense connectivity.

In the literature, the triplet loss tends to utilize the triplets independently without considering the holistic correlation of these triplets, which may result in slow convergence and overfitting of the deep models [33]. To deal with this issue, many efforts have been made to take full advantage of the triplets in training batches [48], [49], [76]–[80]. Specifically, Kumar *et al.* [80] proposed to jointly minimize the triplet loss and a global loss expecting to utilize the global information of the triplets. However, this method optimized the first- and second-order statistics individually and the form of the global loss function is similar to the triplet loss with two additional hyperparameters to be tuned. To overcome this, we develop a novel regularization term inspired by the Fisher discriminant [54] to optimize the statistics jointly, which will lead to more robust performance.

III. Preliminary Knowledge

A. Densely Connected Convolutional Network

As shown in Fig. 2, a typical DenseNet architecture contains an initial convolution followed by repeated cascaded dense block (DB) and transition down (TD) layer combination, and finally a classifier layer is appended. In a DB module, the dense connectivity [40] is expressed as follows:

$$\boldsymbol{x}_l = H_l([\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{l-1}]) \tag{1}$$

where $\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{l-1}$ are the feature maps of all the preceding layers, $[\ldots]$ indicates the concatenation operation, and $\boldsymbol{x}_l$ stands for the lth layer output features with k channels (referred to as the growth rate in [40]). And the composite function H_l represents the four consecutive operations of batch normalization (BN) [81], rectified linear units (ReLUs) [82], a 3×3 convolution, and dropout [83]. The TD layer aims to reduce the size of the feature maps by means of a 1×1 convolution [40] and pooling operation [40]. The last classifier layer makes the final decision using the fully connected (FC) transformation.

Specifically, suppose the input to a DB module has m feature maps and n convolution operations are imposed with each outputting k feature maps, so there will be $(m + n \times k)$ output feature maps of the DB. Compared to the conventional CNNs, the DenseNets can explore many new

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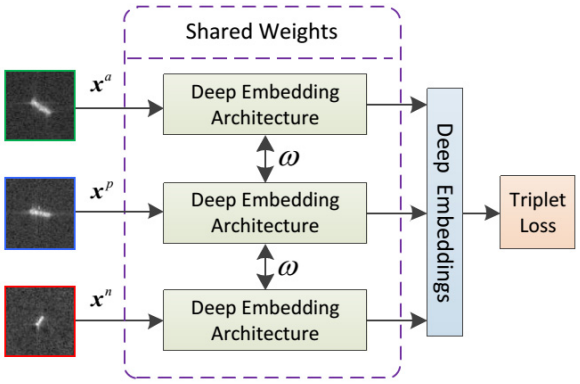


Fig. 3. Model structure of the triplet network with three identical networks sharing the same weights ω. A triplet of (x^a, x^p, x^n) is fed into the deep architecture to obtain the deep embeddings which are then used to compute the triplet loss.

features and require fewer parameters to achieve competing results due to the shortcut connection [40], and the gradient vanishing issue will be alleviated by the direct flow of the loss information to all the preceding layers. All the merits encourage the DenseNets to be applied in various applications [40], [84], [85].

B. Triplet Networks for SAR Image Representation

Taking the SAR images as input, for example, Fig. 3 shows the structure of the triplet networks for discriminative feature learning. Generally, a triplet, comprising an anchor image $\boldsymbol{x}^a$, a positive image $\boldsymbol{x}^p$ from the same class as the anchor and a negative image $\boldsymbol{x}^n$ from another class, is mapped to a d-dimensional Euclidean space by the deep embedding function $f(\boldsymbol{x})$. In the embedding space, the triplets should satisfy the following distance constraint:

$$D(\boldsymbol{x}^a, \boldsymbol{x}^p) + \alpha < D(\boldsymbol{x}^a, \boldsymbol{x}^n) \quad \forall(\boldsymbol{x}^a, \boldsymbol{x}^p, \boldsymbol{x}^n) \tag{2}$$

which enforces the Euclidean distance between the negative paired data ($\boldsymbol{x}^a$, $\boldsymbol{x}^n$) being larger than that of the positive paired data ($\boldsymbol{x}^a$, $\boldsymbol{x}^p$) by a certain margin $\alpha > 0$. The Euclidean distance, e.g., between the positive pair ($\boldsymbol{x}^a$, $\boldsymbol{x}^p$) in the feature space, can be computed as follows:

$$D(\boldsymbol{x}^a, \boldsymbol{x}^p) = \|\boldsymbol{f}(\boldsymbol{x}^a) - \boldsymbol{f}(\boldsymbol{x}^p)\|_2 \tag{3}$$

which is proved effective for metric learning in [78] and [79].

To meet the constraint in (2), the triplet network is optimized by the following triplet loss [33]:

$$L = h(D(\boldsymbol{x}^a, \boldsymbol{x}^p) - D(\boldsymbol{x}^a, \boldsymbol{x}^n) + \alpha) \tag{4}$$

where $h(x) = \max(0, x)$ stands for the hinge loss function [47]. Compared to the contrastive loss [74], the triplet loss can prevent samples of the same class from being projected to a single point in the embedding space [33] and enforce the separability of interclass samples as much as possible, which will naturally benefit the ship classification problems.

IV. Proposed Methodology

A. Proposed Framework of the MR SAR Ship Classification

The common deep models optimized by the softmax loss for classification have achieved impressive results. However, minimizing the softmax loss indicates an equal penalization of the classification error for each class [43] and thus is less effective for dealing with the intraclass variety and interclass similarity. For compensation, referring to Fig. 4, we propose a multitask learning strategy to learn the classification and deep feature embedding, simultaneously, for ship classification in MR SAR images. Specifically, different from the traditional CNN pipeline, some tuples of three images derived from a mini-batch are sent into a triplet network, and then passed through an embedding transformation module to extract the deep feature embeddings. Finally, besides the softmax loss term, a triplet loss term is fused to improve the intraclass compactness and interclass separation. In addition, the weight decay and Fisher discrimination terms are employed as regularization to improve the generalization. In this way, the learned deep features of the same class are much closer to each other and those from different classes are pushed as farther apart as possible. As a result, more discriminative features are extracted for better classification performance.

B. Network Architecture

Inspired by the structure of DenseNet [40], our backbone network is designed as shown in Table I. In a nutshell, the network comprises a stem block, five DBs DB1–DB5, four TD layers TD1–TD4, and an embedding transformation layer which are then followed by the multitask loss. Specifically, to reduce the information loss and inspired in [28], we design a stem layer consisting of two stacked convolutional layers. The convolution operation with a 3×3 filter is denoted by "3×3 conv," each followed by a BN [81] and a ReLU [82] operations, which is the same in other layers. The DB module, referring to Fig. 2(b) and taking DB1, for example, is expressed as "[3×3 conv, 6] $\times$ 4," which contains four convolutions with the growth rate $k = 6$ and filter size of 3×3, likewise the other DB modules. Different from the original DenseNet, the growth rates in the DBs are respectively set to 6, 12, 18, 24, and 30 in an increasing way which is proved effective to explore more new features [86]. In the TD layers, the filter size is set to 3×3 instead of 1×1 to extract more spatial and texture features [87] which will benefit the classification performance. Subsequently, two FC layers are employed to obtain the deep embeddings for the multitask learning.

C. Multitask Learning of Softmax Classification and Deep Feature Embedding

To start with, let $\boldsymbol{X} = \{\boldsymbol{x}_i | i = 1, 2, \ldots, N\}$ denote the training set and $\boldsymbol{Y} = \{\boldsymbol{y}_i | i = 1, 2, \ldots, N\}$ be the set of corresponding labels of $\boldsymbol{X}$, where N is the total number of training samples, $\boldsymbol{y}_i \in \mathbb{R}^C$ is a one-hot vector with the entry representing the true class being 1 and others 0, C is the total number of object classes. For each sample $\boldsymbol{x}_i$, $i = 1, 2, \ldots, N$,

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TABLE I

BACKBONE NETWORK ARCHITECTURE FOR THE MR SAR SHIP CLASSIFICATION. THERE ARE FIVE DBS IN THIS NETWORK. THE INPUT SIZE IS $1 \times 64 \times 64$. ALL THE CONVOLUTION FILTERS TAKE A SMALL BUT EFFECTIVE RECEPTIVE FIELD OF 3×3

Stage	Output Size	Layers
Stem	32×64×64	3×3 conv, stride 1
	32×64×64	3×3 conv, stride 1
DB1	56×64×64	[3×3 conv, 6] × 4
TD1	56×64×64	3×3 conv, stride 1
	56×32×32	2×2 avg_pool, stride 2
DB2	104×32×32	[3×3 conv, 12] × 4
TD2	104×32×32	3×3 conv, stride 1
	104×16×16	2×2 avg_pool, stride 2
DB3	176×16×16	[3×3 conv, 18] × 4
TD3	176×16×16	3×3 conv, stride 1
	176×8×8	2×2 avg_pool, stride 2
DB4	272×8×8	[3×3 conv, 24] × 4
TD4	272×8×8	3×3 conv, stride 1
	272×4×4	2×2 avg_pool, stride 2
DB5	392×4×4	[3×3 conv, 30] × 4
-	392×1×1	4×4 global avg_pool
Embedding Transformation	1000×1	1000-dim FC
	128×1	128-dim FC
	128×1	L2 normalization
Multi-task Loss	-	Softmax + Triplet

also denoted as the anchor sample $\boldsymbol{x}_i^a$, in the training set, a positive sample $\boldsymbol{x}_i^p$ and a negative sample $\boldsymbol{x}_i^n$ can be sampled from the same training set, and the triplet ($\boldsymbol{x}_i^a$, $\boldsymbol{x}_i^p$, $\boldsymbol{x}_i^n$) will thus be obtained as the input to the triplet networks. Assume the designed backbone network has $L+1$ layers, and for simplicity and in spirit of the presentations in [47] and [50], the parameters of the backbone network can be defined as $\mathbf{W} = \{\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_{L+1}\}$ and $\boldsymbol{B} = \{\boldsymbol{B}_1, \boldsymbol{B}_2, \ldots, \boldsymbol{B}_{L+1}\}$, where $\mathbf{W}_l \in \mathbb{R}^{q^{(l)} \times q^{(l-1)}}$ is the filter weight matrix of the lth layer, $l = 1, 2, \ldots, L+1$ and $\boldsymbol{B}_l \in \mathbb{R}^{q^{(l)}}$ is the corresponding biases. For a given input $\boldsymbol{x}_i$, the lth layer output $\boldsymbol{O}_l(\boldsymbol{x}_i)$ can be derived as follows:

$$\boldsymbol{O}_l(\boldsymbol{x}_i) = S_l(\mathbf{W}_l \boldsymbol{O}_{l-1}(\boldsymbol{x}_i) + \boldsymbol{B}_l), \quad l = 1, 2, \ldots, L \tag{5}$$

where S_l is a componentwise nonlinear activation function, i.e., the ReLU [82] function in the convolution layers and the softmax [44] function in the classification layer. Then, the deep embedding feature is derived as follows:

$$\boldsymbol{f}(\boldsymbol{x}_i) = \frac{\boldsymbol{O}_L(\boldsymbol{x}_i)}{\|\boldsymbol{O}_L(\boldsymbol{x}_i)\|_2}. \tag{6}$$

The L2-norm normalization is to enforce the deep features to locate in a unit hypersphere in the high dimensional embedding space to eliminate the scale effect [50].

1) Softmax Classification: Accordingly, the output of the Softmax classifier layer is defined as follows:

$$\boldsymbol{O}_{L+1}(\boldsymbol{x}_i) = S_{L+1}(\mathbf{W}_{L+1} \boldsymbol{f}(\boldsymbol{x}_i) + \boldsymbol{B}_{L+1}) \tag{7}$$

where the entry in $\boldsymbol{O}_{L+1}(\boldsymbol{x}_i)$ indicates the posterior probability of $\boldsymbol{x}_i$ being classified into the corresponding class. And thus the softmax log-loss can be expressed as follows:

$$J_s(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} \boldsymbol{y}_i^c \log \boldsymbol{O}_{L+1}^c(\boldsymbol{x}_i). \tag{8}$$

2) Deep Feature Embedding Learning: The softmax log-loss penalizes the classification error equally for each class which cannot enforce the learned deep features discriminative enough. Thus, inspired by the metric learning method, we apply the triplet loss in (4) as the similarity constraint to pull the positive sample pairs in the embedding space much closer to each other and push the negative sample pairs far apart from each other. Specifically, the triplet loss for the training batch with N triplets will be described as follows:

$$J_t\big(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}\big) = \frac{1}{N} \sum_{i=1}^{N} h\big(D\big(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\big) - D\big(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\big) + \alpha\big). \tag{9}$$

The learned deep embeddings using the triplet loss above will have small intraclass variance and large interclass separation, which are discriminative enough for better classification.

3) Multitask Framework: To incorporate the advantages of the softmax classification and DML, we apply a multitask framework to learn more discriminative features for MR SAR ship classification. The framework is formulated by integrating the two losses above as follows:

$$J = \lambda J_s(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) + (1 - \lambda) J_t(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) \tag{10}$$

where $0 < \lambda < 1$ is the weight to tradeoff the relative importance of the two loss terms. In practice, λ is high enough, e.g., larger than 0.5, to lay more stress on the supervised classification information.

Benefited from the multitask framework, the deep models are jointly optimized by propagating the classification error information and the similarity constraint information backward simultaneously. In such a way, more powerful deep features suitable for classification are obtained.

4) Regularization Terms: To train the deep neural networks robustly, some regularization techniques need to be adopted. The commonly used one is the weight decay [50] term. This term imposes an L2-norm or the Frobenius norm penalty on the learnable parameters of the networks that is defined as follows:

$$J_w(\mathbf{W}, \boldsymbol{B}) = \sum_{l=1}^{L+1} \big(\|\mathbf{W}_l\|_F^2 + \|\boldsymbol{B}_l\|_2^2\big). \tag{11}$$

By adding the weight decay term to the objective function, large values of the network parameters are penalized, which is beneficial to prevent over-fitting.

Despite the prominent performance of the triplet loss in image retrieval and face recognition, however, there are still some defects for which the triplet loss is blamed. For instance, given a training batch with N samples, there are about $O(N^3)$ possible triplets. When N is large enough, it is infeasible for

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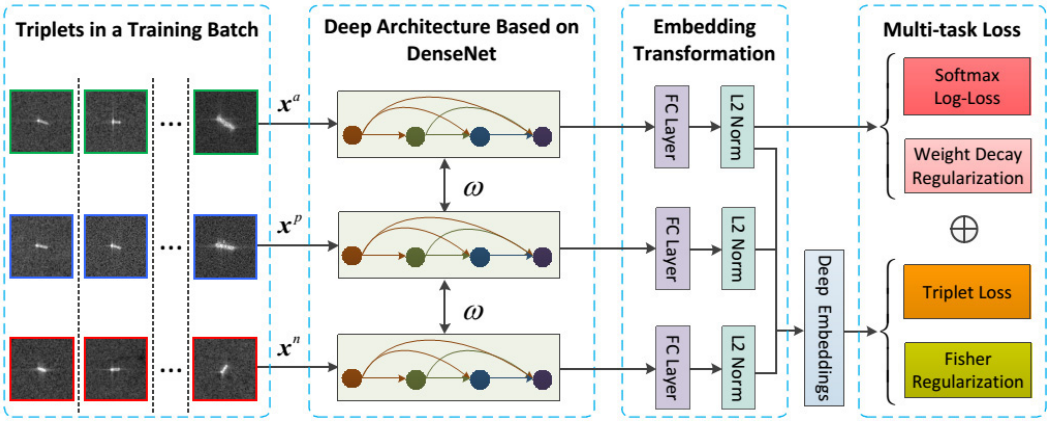


Fig. 4. Flowchart of the proposed MR SAR ship classification method. A set of triplets in a training batch are fed into the triplet network and a sequent embedding transformation to extract deep embeddings. For multitask learning, a triplet loss is combined with the softmax loss to learn more discriminative deep features. To fully mine the relationships between the triplets, a Fisher regularization term is imposed on the deep features to explore the contextual information for better classification performance.

training the networks. Additionally, as the training goes, more triplets will obey the constraint (2), which means these triplets have no contribution for parameter updating and results in slow convergence [33]. On the other hand, the fixed margin in (9) is inflexible for training. If the margin is too small, the training process is less sound to sufficiently utilize the triplets. And if the margin is too large, the training process will focus mainly on the very hard triplets, which will easily lead to over-fitting and model collapse [33], [77]–[79]. Both of them can impair the performance of the learned models. In the recent literature, many researchers commonly embarked on the triplet sampling and loss function design to improve the potentials of the deep models [76]–[80]. In this article, we employ an online triplet sampling method for training the deep models which will be introduced in Section V-A. And most importantly, as shown in the following, we apply the Fisher discrimination criterion [54] to incorporate the contextual information in a training batch and mitigate the over-fitting induced by the fixed margin in (9).

As pointed out in [78] and [80], the pairwise distances of the deep embeddings from the same or different classes follow two distinct distributions (e.g., the Gaussian distribution). And thus, inspired by the Fisher discrimination, we propose to regularize the triplet loss by formulating the following term:

$$J_f(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) = \frac{\sigma_1^2 + \sigma_2^2}{(m_1 - m_2)^2} \tag{12}$$

where m_1 and m_2 are, respectively, the mean Euclidean distances of the positive and negative pairs in the feature space and σ_1^2 and σ_2^2 are the corresponding variances. These statistics are computed as follows:

$$\begin{cases} m_1 = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) \\ m_2 = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\right) \end{cases} \tag{13}$$

$$\begin{cases} \sigma_1^2 = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \left(D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) - m_1\right)^2 \\ \sigma_2^2 = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \left(D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\right) - m_2\right)^2 \end{cases} \tag{14}$$

where N is the total number of triplets in a training batch. Different from [80] and [88], we formulate the Fisher regularization by integrating the first- and second-order statistics together in the form of the inverse Fisher discriminant. So, compared to the global loss in [80], this proposed regularization term has less hyperparameters to be tuned and may achieve better performance benefited from the joint optimization. Additionally, one can find that the contextual information is fully investigated in (12) such that the downsides of the triplet loss are mitigated and it is thus to ensure the robust training of the triplet networks.

To summarize, as shown in Fig. 4, by incorporating the terms (11) and (12) into (10), the final objective function can be formulated as follows:

$$\begin{aligned} J &= \lambda_1 J_s(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) + (1-\lambda_1) J_t(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) \\ &\quad + \lambda_2 J_f(\boldsymbol{X}, \mathbf{W}, \boldsymbol{B}) + \frac{\lambda_3}{2} J_w(\mathbf{W}, \boldsymbol{B}) \\ &= -\frac{\lambda_1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} \boldsymbol{y}_i^c \log \boldsymbol{O}_{L+1}^c(\boldsymbol{x}_i) \\ &\quad + \frac{1-\lambda_1}{N}\sum_{i=1}^{N} h\left(D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) - D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\right) + \alpha\right) \\ &\quad + \lambda_2 \frac{\sigma_1^2 + \sigma_2^2}{(m_1 - m_2)^2} + \frac{\lambda_3}{2}\sum_{l=1}^{L+1}\left(\|\mathbf{W}_l\|_F^2 + \|\boldsymbol{B}_l\|_2^2\right). \end{aligned} \tag{15}$$

By minimizing the objective function (15), the parameters of the networks are obtained which will be used for inference.

D. Optimization

In this article, we utilize the stochastic gradient descent (SGD) algorithm [89] to solve the minimization problem of the objective function in (15). Referring to the presentations in Section IV-C, the parameters in the $(L+1)$th layer are related to the classification error and the weight decay term, and the parameters of all the first L layers are related to the holistic terms in (15) during back-propagation [89], so the gradients can be efficiently computed in an iterative way. To be specific, to acquire the network parameters $\{\mathbf{W}_l, \boldsymbol{B}_l\}$, $l = 1, 2, \ldots, L+1$, the gradients of the objective function

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J with regard to the parameters $\{\mathbf{W}_{L+1}, \boldsymbol{B}_{L+1}\}$ and $\{\mathbf{W}_l, \boldsymbol{B}_l\}$, $l = 1, 2, \ldots, L$, can be, respectively, calculated as follows:

$$\frac{\partial J}{\partial \mathbf{W}_{L+1}} = \frac{\lambda_1}{N}\sum_{i=1}^{N}\delta_{L+1}\boldsymbol{O}_L(\boldsymbol{x}_i)^{\mathrm{T}} + \lambda_3\mathbf{W}_{L+1} \tag{16}$$

$$\frac{\partial J}{\partial \boldsymbol{B}_{L+1}} = \frac{\lambda_1}{N}\sum_{i=1}^{N}\delta_{L+1} + \lambda_3\boldsymbol{B}_{L+1} \tag{17}$$

$$\begin{aligned}\frac{\partial J}{\partial \mathbf{W}_l} = {} & \frac{\lambda_1}{N}\sum_{i=1}^{N}\delta_l\boldsymbol{O}_{l-1}(\boldsymbol{x}_i)^{\mathrm{T}} + \frac{1-\lambda_1}{N} \\ & \times \sum_{i=1}^{N}\left(\delta_l^{ha}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^a\right)^{\mathrm{T}} + \delta_l^{hp}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^p\right)^{\mathrm{T}}\right. \\ & \left. - \delta_l^{hn}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^n\right)^{\mathrm{T}}\right) \\ & + \lambda_2\sum_{i=1}^{N}\left(\delta_l^{ga}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^a\right)^{\mathrm{T}} + \delta_l^{gp}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^p\right)^{\mathrm{T}}\right. \\ & \left. + \delta_l^{gn}\boldsymbol{O}_{l-1}\left(\boldsymbol{x}_i^n\right)^{\mathrm{T}}\right) \\ & + \lambda_3\mathbf{W}_{L+1}\end{aligned} \tag{18}$$

$$\begin{aligned}\frac{\partial J}{\partial \boldsymbol{B}_l} = {} & \frac{\lambda_1}{N}\sum_{i=1}^{N}\delta_l + \frac{1-\lambda_1}{N}\sum_{i=1}^{N}\left(\delta_l^{ha} + \delta_l^{hp} - \delta_l^{hn}\right) \\ & + \lambda_2\sum_{i=1}^{N}\left(\delta_l^{ga} + \delta_l^{gp} + \delta_l^{gn}\right) + \lambda_3\boldsymbol{B}_{L+1}\end{aligned} \tag{19}$$

where $\boldsymbol{O}_0(\boldsymbol{x}_i) = \boldsymbol{x}_i$ and $\boldsymbol{O}_0(\boldsymbol{x}_i^r) = \boldsymbol{x}_i^r$, $r = a,\ p,\ n$ are the inputs of the triplet networks. In our triplet sampling method, the anchor sample $\boldsymbol{x}_i^a$ corresponds to each sample $\boldsymbol{x}_i$ in the training set. And the updating equations are defined as follows:

$$\delta_{L+1} = \frac{\boldsymbol{O}_{L+1}(\boldsymbol{x}_i) - \boldsymbol{y}_i}{\|\boldsymbol{O}_L(\boldsymbol{x}_i)\|_2} \tag{20}$$

$$\delta_L = \boldsymbol{Z}(\boldsymbol{u}_L(\boldsymbol{x}_i))\mathbf{W}_{L+1}^{\mathrm{T}}\delta_{L+1} \odot S_L'(\boldsymbol{u}_L(\boldsymbol{x}_i)) \tag{21}$$

$$\delta_l = \mathbf{W}_{l+1}^{\mathrm{T}}\delta_{l+1} \odot S_l'(\boldsymbol{u}_l(\boldsymbol{x}_i)), \quad l = 1, 2, \ldots, L-1 \tag{22}$$

$$\begin{aligned}\delta_L^{ha} = {} & h'(\upsilon)\boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^a\right)\right)\left(\frac{\boldsymbol{f}(\boldsymbol{x}_i^a) - \boldsymbol{f}(\boldsymbol{x}_i^p)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^p\right)\right\|_2}\right. \\ & \left. - \frac{\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^n\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^n\right)\right\|_2}\right) \\ & \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^a\right)\right)\end{aligned} \tag{23}$$

$$\delta_L^{hp} = h'(\upsilon)\boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^p\right)\right)\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^p\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^p\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)\right\|_2} \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^p\right)\right) \tag{24}$$

$$\delta_L^{hn} = h'(\upsilon)\boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^n\right)\right)\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^n\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^n\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)\right\|_2} \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^n\right)\right) \tag{25}$$

$$\begin{aligned}\delta_l^{hr} = {} & \mathbf{W}_{l+1}^{\mathrm{T}}\delta_{l+1}^{hr} \odot S_l'\left(\boldsymbol{u}_l\left(\boldsymbol{x}_i^r\right)\right) \\ & r = a,\ p, n, l = 1, 2, \ldots, L-1\end{aligned} \tag{26}$$

$$\begin{aligned}\delta_L^{ga} = {} & \boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^a\right)\right)\left(g_{D_{ap}}'\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^p\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^p\right)\right\|_2}\right. \\ & \left. - g_{D_{an}}'\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^n\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^n\right)\right\|_2}\right) \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^a\right)\right)\end{aligned} \tag{27}$$

$$\delta_L^{gp} = \boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^p\right)\right)g_{D_{ap}}'\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^p\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^p\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)\right\|_2} \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^p\right)\right) \tag{28}$$

$$\delta_L^{gn} = \boldsymbol{Z}\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^n\right)\right)g_{D_{an}}'\frac{\boldsymbol{f}\left(\boldsymbol{x}_i^n\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)}{\left\|\boldsymbol{f}\left(\boldsymbol{x}_i^n\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^a\right)\right\|_2} \odot S_L'\left(\boldsymbol{u}_L\left(\boldsymbol{x}_i^n\right)\right) \tag{29}$$

$$\begin{aligned}\delta_l^{gr} = {} & \mathbf{W}_{l+1}^{\mathrm{T}}\delta_{l+1}^{gr} \odot S_l'\left(\boldsymbol{u}_l\left(\boldsymbol{x}_i^r\right)\right) \\ & r = a, p, n, l = 1, 2, \ldots, L-1\end{aligned} \tag{30}$$

where the operation $\odot$ denotes the element-wise multiplication, and the intermediate notations $\boldsymbol{Z}$, $\boldsymbol{u}$, υ, and g in (21)–(30) are defined as follows:

$$\boldsymbol{Z}(\boldsymbol{u}_L(\boldsymbol{x}_i)) \triangleq \frac{\mathbf{I}}{\|\boldsymbol{O}_L(\boldsymbol{x}_i)\|_2} - \frac{\boldsymbol{O}_L(\boldsymbol{x}_i)\boldsymbol{O}_L(\boldsymbol{x}_i)^{\mathrm{T}}}{\|\boldsymbol{O}_L(\boldsymbol{x}_i)\|_2^3} \tag{31}$$

$$\boldsymbol{u}_l(\boldsymbol{x}_i) \triangleq \mathbf{W}_l\boldsymbol{O}_{l-1}(\boldsymbol{x}_i) + \boldsymbol{B}_l, \quad l = 1, 2, \ldots, L \tag{32}$$

$$\upsilon \triangleq D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) - D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\right) + \alpha \tag{33}$$

$$g \triangleq \frac{\sigma_1^2 + \sigma_2^2}{(m_1 - m_2)^2} \tag{34}$$

where $\mathbf{I}$ is the identity matrix. $h'(\upsilon)$ and $g_{D_{ar}}'$ are the derivatives with regard to the respective variables, which are defined as follows:

$$h'(\upsilon) \triangleq \frac{\partial h(\upsilon)}{\partial \upsilon} \tag{35}$$

$$g_{D_{ar}}' \triangleq \frac{\partial g}{\partial D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^r\right)}, \quad r = p, n. \tag{36}$$

Similarly, $S_l'(\cdot), l = 1, 2, \ldots, L$ is the derivative of the nonlinear activation function which is then applied componentwise.

Then, the network parameters $\{\mathbf{W}_l, \boldsymbol{B}_l\}$, $l = 1, 2, \ldots, L+1$, can be updated iteratively by using the gradient descent algorithm as follows until convergence:

$$\mathbf{W}_l = \mathbf{W}_l - \mu\frac{\partial J}{\partial \mathbf{W}_l} \tag{37}$$

$$\boldsymbol{B}_l = \boldsymbol{B}_l - \mu\frac{\partial J}{\partial \boldsymbol{B}_l} \tag{38}$$

where μ is the learning rate.

V. Implementation Details

A. Triplet Sampling

As shown in [33], it is crucial to select appropriate triplets for network training. If the triplets selected satisfy the constraint (2) easily, it will lead to slow convergence [33]. However, to mine the hardest triplets across the whole data set, it might be trapped in bad local minima. To make a tradeoff, in our work, an online method that is based on [90] is employed for triplet selection.

Specifically, suppose there are C classes with K samples per class in a training batch. For each sample in one training batch being selected as an anchor image, which is denoted by $\{\boldsymbol{x}_i^a\}_{i=1,\ldots,C}^{a=1,\ldots,K}$, and then correspondingly, the hardest positive sample $\boldsymbol{x}_i^p$ and the hardest negative sample $\boldsymbol{x}_j^n$ can be,

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respectively, sampled as follows:

$$\begin{aligned} \boldsymbol{x}_i^p &= \arg\max_{\boldsymbol{x}_i^p} D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) \\ &= \left\| \boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_i^p\right) \right\|_2 \quad \forall p = 1, \ldots, K \end{aligned} \tag{39}$$

$$\begin{aligned} \boldsymbol{x}_j^n &= \arg\min_{\boldsymbol{x}_j^n} D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_j^n\right) \\ &= \left\| \boldsymbol{f}\left(\boldsymbol{x}_i^a\right) - \boldsymbol{f}\left(\boldsymbol{x}_j^n\right) \right\|_2 \quad \forall j = 1, \ldots, C, \quad j \neq i, \ n = 1, \ldots, K. \end{aligned} \tag{40}$$

Thus, for a training batch with N samples, we can totally get N triplets. This triplet sampling method is called Batch Hard in [90], which is proved effective in our experiment.

B. Activation Function and Initialization

To improve the representation ability of the deep models, nonlinear activation function is always applied. In our designed CNN networks, the ReLU [82] function is applied to all the entries of the output feature maps of each convolutional layer element-wisely. This nonsaturating function is defined as follows:

$$S(x) = \max(0, x). \tag{41}$$

And its derivative is computed as follows:

$$S'(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0. \end{cases} \tag{42}$$

For training deep CNNs, it is important to avoid reducing or magnifying the magnitudes of the signals exponentially during forward or backward propagation [91]. And thus we utilize the weight initialization method in [91], within which the weights of each convolutional layer are initialized by the following Gaussian distribution:

$$\mathbf{W}_l \sim N(0, \sqrt{2/n_{\text{in}}}) \tag{43}$$

where n_{in} is the number of connections of a response in the lth layer. And n_{in} is computed as [91]

$$n_{\text{in}} = k_l^2 d_{l-1} \tag{44}$$

where k_l is the filter size of the lth layer and d_{l-1} is the number of feature maps of the $(l-1)$th layer. For better performance, the weights of the FC layers are initialized by the following uniform distribution [92]:

$$\mathbf{W} \sim U\left[-\frac{\sqrt{6}}{\sqrt{n_{\text{in}} + n_{\text{out}}}}, \frac{\sqrt{6}}{\sqrt{n_{\text{in}} + n_{\text{out}}}} \right] \tag{45}$$

where n_{in} and n_{out} are the numbers of the input and output neurons of FC layers, respectively. The biases of the networks are all initialized to zeros.

C. Inference

As the triplet networks have been trained, we only need one branch for inference. That is to say, the triplet loss is removed during testing. And the test samples are directly fed to the backbone network trained to extract the deep embedding features. Then, the Softmax classifier is used to predict the specific category of each test sample.

TABLE II
STATISTICS OF THE REARRANGED SAR DATA SET

Ship Type	Tanker	Container ship	Bulk carrier	Cargo	General cargo
VH	360	217	802	977	121
VV	360	217	802	977	121
Total	720	434	1604	1954	242

VI. EXPERIMENTS

In this section, we will perform extensive experiments to verify the effectiveness of the proposed method. First, we present the experimental settings and evaluation metrics, also including the data set acquisition based on the OpenSARShip [18] database and data set setting. Then, the hyperparameter selection, validation of the Fisher regularization, and ablation study are conducted. Finally, we perform a comprehensive comparison between the proposed method and the state-of-the-art methods as well as the classical CNNs for ship classification in MR SAR images. In addition, all the experiments are implemented on a DELL workstation with the Intel Xeon Gold 6128 CPU (6 cores, 3.4 GHz), the NVIDIA GeForce GTX 1080 Ti GPU (11 GB) and a 128-GB memory. And the network training and testing are conducted using the TensorFlow (ver. 1.10) [93] deep learning framework distributed by Google in the Ubuntu 16.04 environment.

A. Data Set Preparation

Thanks to the release of the OpenSARShip [18] database, it is possible to implement ship classification across large-scale SAR data set. However, there are some drawbacks for this data set. First, the sizes of the ship chips are different for various types, which is cumbersome to be applied to the normal CNNs. And for some ship chips, more than two targets exist in one chip that might impair the recognition performance. Most importantly, to some extent, the classified ship types are a bit coarse. For example, the common merchant vessels, i.e., bulk carriers, container ships, and general cargos, are all denoted as the cargo ships, in which it is infeasible to perform fine-grained ship classification. To accommodate these issues, we propose to rearrange the OpenSARShip data set to acquire a friendly data set for fine-grained ship classification, as shown in Fig. 5.

The OpenSARShip data set [18] was collected from the Sentinel-1 SAR images[1] with IW mode and VH and VV dual polarizations. Specifically, we exploit the ground range detected (GRD) products in VH–VV polarizations owing to the equal pixel spacing (10 m) in both the range and azimuth [18], which can preserve the physical properties of the ships. Following the procedure in Fig. 5, we obtain the ship chips for five common merchant vessels, i.e., tanker, container, bulk carrier, cargo, and general cargo. The numbers of each ship type are listed in Table II. And some hard-to-be-classified samples of the five merchant ship types are illustrated in Fig. 6. As commonly practiced, the classification of three merchant vessels, tanker, container, and bulk carrier, is carried out, due

[1] https://sentinels.copernicus.eu/web/sentinel/sentinel-data-access

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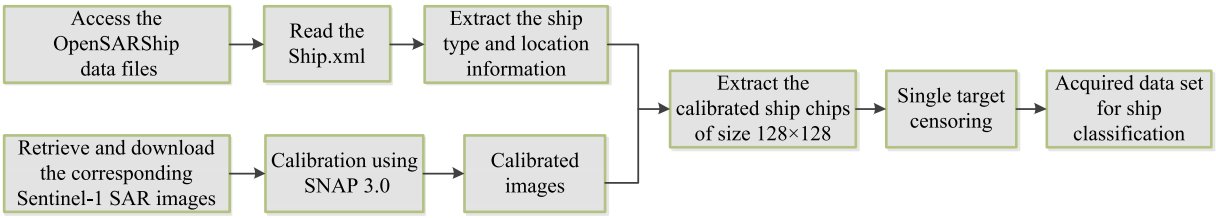


Fig. 5. Procedure of the SAR ship data set acquisition. First, the access to the OpenSARShip [18] data files is required to obtain the ground truth information of the ship chips. Second, the corresponding Sentinel-1 SAR images are downloaded from the official website and are calibrated by the SNAP 3.0 software [18]. Third, the equal-sized ship chips with specific categories are extracted based on the calibrated images and the extracted ship type and location information. As postprocessing and to obtain the single target chips, the interfering targets and clutter are replaced by the surrounding sea clutters. And finally, the novel rearranged data set used for our experiment is obtained. Some samples of the ship chips are shown in Fig. 6.

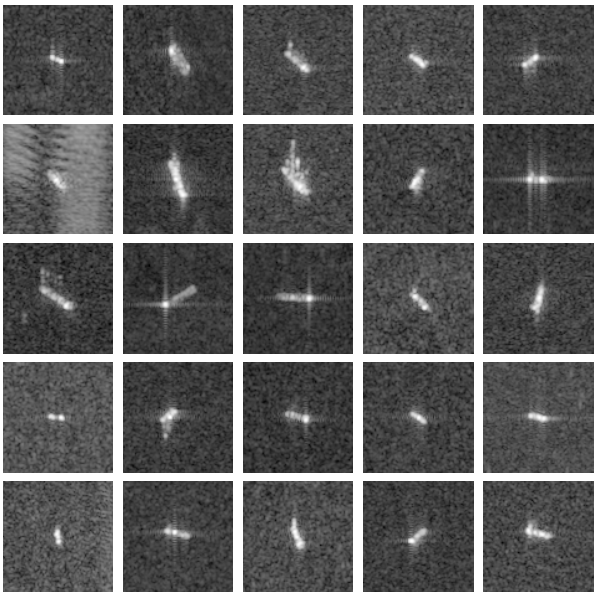


Fig. 6. Illustration of some hard-to-be-classified ship samples of (first row) the tanker, (second row) container ship, (third row) bulk carrier, (fourth row) cargo, and (fifth row) general cargo. Some interference, such as sidelobes, azimuth smearing [18], and strong sea clutter, will degenerate the recognition performance.

to their large amount among the international shipping market [59], [61]. The validation experiments are mainly performed based on this task. After validation, the training and validation data sets are jointly used to retrain the network from scratch and the obtained models are employed for testing. To further verify the robustness of the proposed method, we also extend the experiments to classify the five ship types and provide some quantitative results.

B. Data Set Setting and Evaluation

In our experiments, we use both the VH and VV samples as processed in [61], [62], and [70]. We divide the entire data set into training data set, validation data set, and test data set. The split ratio between each part is about 7:1:2 and the statistics of the data set split are listed in Table III. To improve the reliability of the experimental results, as similar to [30], [33], [50], and [51], we conduct five rounds of the data set split by randomly selecting the samples for each subset. The final performance is reported by the mean accuracy and standard error of the five rounds. After validation, the training data set and the validation data set are jointly used to train the networks.

TABLE III
STATISTICS OF THE REARRANGED SAR DATA SET SPLIT

Ship Type	Training	Validation	Testing	Total
Tanker	490	80	150	720
Container ship	304	50	80	434
Bulk carrier	1064	170	370	1604
Cargo	1504	-	450	1954
General cargo	192	-	50	242
Total	3554	300	1100	4954

To train the CNNs, the amount of data set is very important. And the data volume of each ship type in our data set is unbalanced. Thus, for better network training, we propose to implement data augmentation for the training data set, as in [70], [80], and [94], and the transformations are as follows.

1) Horizontally and vertically flip the ship chips.
2) Rotate the ship chips by 90°, 180°, and 270°.
3) Randomly translate the ship chips with the translation value of pixels selected from -5 to 5.
4) Add the Gaussian noise with the mean and standard deviation set as 0 and 0.001, respectively.

After augmentation, the amount of ship chips of each type is 8 times larger than the initial number, i.e., containing the 7 times augmented samples and the initial ones. To acquire equal number of samples for each ship type (confined by the ship type with least samples), the final training data set contains the initial samples and the ones randomly selected from the 7 times augmented samples. Note that all the augmentation operations are directly applied to the GRD magnitude images, and some technical data augmentation methods considering the SAR imaging characteristics should be further explored in the future.

Additionally, to improve the classification accuracy and reduce the computation cost, the ship chips for experiment are cropped from the center with size of 64×64. And for better performance, as in [67] and [87], the ship chips are normalized by respective mean values and standard deviations as follows:

$$\hat{x} = \frac{x - \mu_x}{\sigma_x} \tag{46}$$

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TABLE IV

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 3-CLASS VALIDATION DATA SET USING DIFFERENT TRIPLET LOSS TRADEOFF PARAMETERS

λ_1	0.6	0.7	0.8	0.9	1.0
Accuracy	**87.67 ± 0.78**	87.33 ± 0.67	87.53 ± 1.26	87.66 ± 1.25	87.20 ± 1.74

where x and $\hat{x}$ are the original and normalized pixel values of a ship chip, respectively. μ_x and σ_x are the corresponding mean and standard deviation.

For evaluation, the overall accuracy and confusion matrix are employed. The overall accuracy is a whole indication of the classification performance by calculating the proportion of the correctly classified samples in the test data set. The confusion matrix is an informative table, in which the diagonal entries are the correct classification accuracies for each ship type and the nondiagonal entries are the confusions indicating the ratios of the ship types in the rows misclassified into the ship types in the columns. To implement, the confusion matrix is computed from the experiment whose overall accuracy is the median value of the five runs.

C. Parameter Setting

The proposed networks are trained by the SGD algorithm. Initially, the learning rate, the momentum [27], the weight decay parameter [i.e., λ_3 in (15)], and the batch size are empirically set to 0.01, 0.9, 0.0005, and 100, respectively, which are appropriate for our method. To alleviate overfitting, the dropout rate is set as 0.2. For the triplet margin in (9), too large value will lead to overfitting, so we properly set this parameter to 0.2. The networks are totally trained for 300 epochs, and the learning rate is divided by 10 when the epoch approaches 150, 200, and 250, respectively. It is worth to note that, during initial optimization, the learned deep features in embedding space are less informative, so the optimization of the Fisher regularization term is kind of hard. To leverage the training, we add the triplet loss and the Fisher regularization term for optimization after 10 epochs of training using the softmax loss solely. In the following, particularly, we take a sensitivity analysis on the parameters λ_1 and λ_2 in (15) and the embedding dimensions.

1) Test the Triplet Loss Term: To test the effectiveness of the triplet similarity constraint, we employ the softmax loss term and the triplet loss term in (15) to train the networks. The values of the parameter λ_1 are selected from the set of {0.6, 0.7, 0.8, 0.9, 1.0}. For each value, the experiment is conducted 5 times on the five splits of training data set and validation data set. The results are reported by the mean accuracies and the standard deviations, which are shown in Table IV. From the results in Table IV, one can observe that the triplet similarity constraint has a positive impact on the performance improvement of the ship classification. And different values in [0.6, 0.9] have a small performance difference (within 0.4%). To leverage more triplet similarity constraint, hence, we fix the tradeoff parameter λ_1 to 0.6 in the following.

TABLE V

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 3-CLASS VALIDATION DATA SET USING DIFFERENT FISHER REGULARIZATION TRADEOFF PARAMETERS

λ_2	0.001	0.005	0.01	0.1
Accuracy	87.67 ± 0.78	**88.13 ± 0.65**	87.73 ± 1.96	87.47 ± 1.17

TABLE VI

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 3-CLASS VALIDATION DATA SET USING DIFFERENT VALUES OF EMBEDDING DIMENSIONALITY

Dim.	64	128	256	512
Accuracy	87.53 ± 0.56	**88.13 ± 0.65**	87.80 ± 0.84	87.27 ± 0.55

2) Test the Fisher Regularization Term: To verify the effect of the Fisher regularization term, we employ the full version of (15) to train the networks. The tradeoff parameter λ_2 is set in the range of {0.001, 0.005, 0.01, 0.1}. The results are reported in Table V, from which some conclusions can be drawn. That is to say, a proper tradeoff parameter λ_2 is required to balance the Fisher regularization constraint. If this parameter is a bit small, less global information is contained in a training batch to perfectly optimize the network parameters. While this parameter is too large, e.g., taking the value 0.1, too much attention is paid on this term. In other words, much more efforts, during the model training, are made to contain the global information excessively, which will enforce the learned models to be more sensitive to the data noises and thus the model performance may decrease. In our experiments, the tradeoff parameter λ_2 is fixed to 0.005 as shown in Table V.

3) Test the Embedding Dimensionality: We vary the values of the embedding dimensionality to evaluate the representation ability. The dimensional values of the embeddings are selected from the set of {64,128, 256, 512} and the results are shown in Table VI. One can find that, if the embedding dimensionality is too small or much larger, the model accuracy will degrade. It is straightforward that the representational ability is weak for low-dimensionality embeddings. And if the dimensionality is too large, overfitting may emerge over such finite-scale data set. Hence, in the following, the embedding dimensionality is fixed to 128 for all the experiments.

D. Verification of the Multitask Learning

In this section, we take a further step to verify the efficacy of the triplet loss and the Fisher regularization term. For evaluation, we compute the difference values of the triplet distances as the similarity constraint in (2) and the Fisher discriminant, which are respectively defined as follows:

$$\text{Diff} = \frac{1}{N}\sum_{i=1}^{N}\left(D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^p\right) - D\left(\boldsymbol{x}_i^a, \boldsymbol{x}_i^n\right)\right) \tag{47}$$

$$\text{FD} = \frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2}. \tag{48}$$

The two evaluation metrics are computed in a training batch as aforementioned, and the corresponding value for each epoch is obtained by averaging the corresponding metrics over all the

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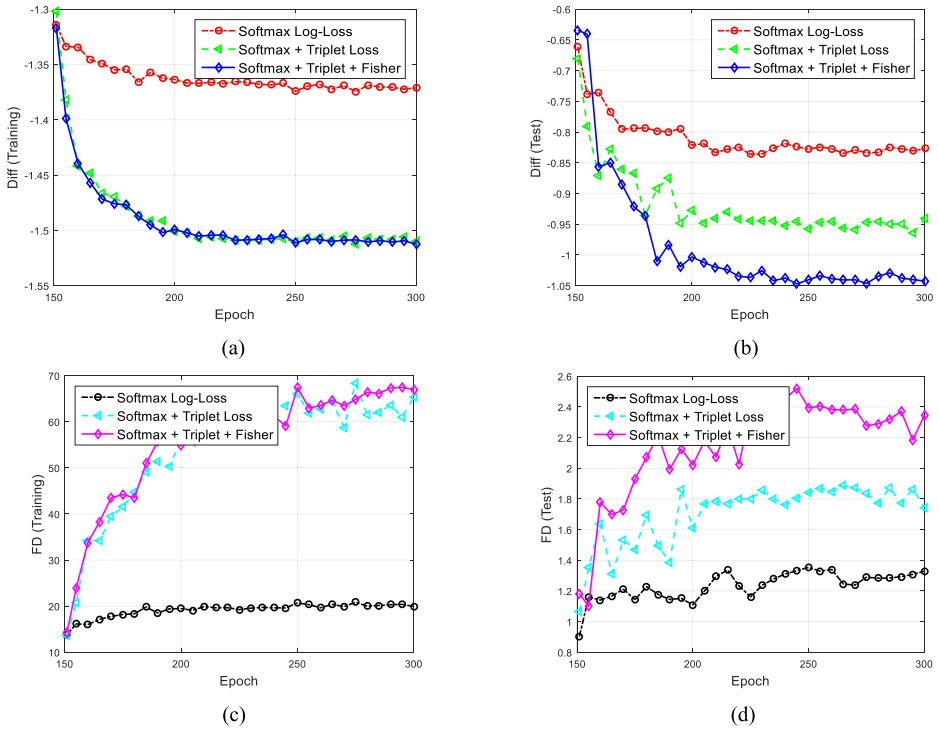


Fig. 7. Testify the effectiveness of the proposed multitask learning framework. In each plot, the evaluation curves are illustrated over three different optimization objective combinations, i.e., "Softmax Log-Loss," "Softmax + Triplet Loss," and "Softmax + Triplet + Fisher." The curves of Diff values versus the epoch number are shown in (a) and (b) for the training and test data sets, respectively. So does the case for the curves of FD values versus the epoch number in (c) and (d).

batches. For better illustration and inspect the performance improvement detailedly, in Fig. 7, we show the curves of the Diff and FD versus the epoch number which varies from 151 to 300. It is straightforward that the smaller values of the Diff mean the positive samples are much further from the negative samples in the embedding space and larger FD values indicate the larger interclass scatter and smaller intraclass variances. To inspect the varying curves in Fig. 7, we can draw the following conclusions.

1) During the training process, for both of the training and test data sets, the Diff values keep consistently decreasing and the FD values become much larger. This implies that different kinds of ship types tend to be separated much further in the learned embedding space and the samples from the same category are enforced to be more compact and those from different classes are pushed far away from each other. Therefore, to some extent, diversified ship samples can be sufficiently distinguished as the network converged.
2) Comparing with the softmax log-loss, the addition of the triplet loss and the Fisher regularization term can acquire smaller Diff values and larger FD values. This is rational that, in addition to the softmax classification error penalty, the triplet loss strives to explore the pairwise distance relations of enormous triplets and the Fisher regularization term can capture the context information in a training batch. It is worth to note that, for the training data set, the evaluation metrics for the "Softmax + Triplet Loss" and "Softmax + Triplet + Fisher" optimization combinations have small margins. This might be due to the fact that, as the training process goes, the network parameters learned can sufficiently fit the training data. In fact, during the last 100 epochs, the overall accuracy of the training data will approach up to above 99.9%. Hence, the network performance is too well to demonstrate the potential of the multitask learning on the training data set.
3) Comparing the evaluation curves between the training and test data sets, one can find that the performance margins among different optimization objective combinations are much larger for the test data set. This indicates that, within the multitask learning framework, the network models learned with the triplet loss and Fisher regularization term have prominent generalization abilities.

To summarize, the triplet loss can bring the ship samples of the same class closer and push the ones of different classes far apart in the embedding space. And the Fisher regularization can improve the robustness of the networks by comprising the contextual information. Thus, the designed multitask learning framework is effective for the MR SAR ship classification.

E. Ablation Study

In this section, we perform the ablation study for different optimization objective combinations of the proposed multitask learning framework on the test data set. All the network parameters are set as the default values according to the parameter setting experiments.

Experimental results are shown in Table VII, one can find that the classification accuracies achieved by distinct objective

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TABLE VII
OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 3-CLASS TEST DATA SET USING DIFFERENT OBJECTIVE FUNCTION COMBINATIONS

Objective function combination	Accuracy
Softmax Log-Loss	87.73 ± 0.42
Softmax + Triplet Loss	88.27 ± 0.56
Softmax + Triplet + Fisher	$\mathbf{88.97 \pm 0.72}$

function combinations are promising, which are up to as high as 88.97% on the 3-class MR SAR ship classification task. Compared with the baseline network optimized by the softmax log-loss solely, the joint optimizations of the "Softmax + Triplet Loss" as well as the "Softmax + Triplet + Fisher" could obtain better results, which decreases the initial classification error (i.e., 12.27%) by a relative percentage of 4.4% and 10.11%, respectively, with regard to the baseline method. Hence, the integration of the triplet similarity constraint and the Fisher regularization is crucial to learn more discriminative deep features for better ship classification performance.

F. Extensive Comparisons

To comprehensively evaluate the performance improvement of the proposed method, we conduct extensive experiments by comparing the proposed method with the recent state-of-the-art CNN methods, i.e., the CNN-A [66], CNN-B [37], CNN-C [38], and CNN-D [36], which are also summarized in [70], and with the classical CNNs such as the AlexNet [27], VGGNet [28], GoogLeNet [29], and ResNet [30]. For brevity and space limitation, the structures of the state-of-the-art CNNs are presented in Table VIII, if necessary, with a slight modification to fit the input size of our data set and the structures of the classical CNNs can be referred to their respective references. It should be noted that, to avoid overfitting, the dimensionality of the penultimate FC layer in the classical CNNs is all set to 128 and the usual VGGNet-16 [28] and ResNet-18 [30] are employed. All the comparing CNNs are trained from scratch on the collected SAR ship data. The proposed backbone network in Table I optimized by softmax log-loss solely is simply denoted as SAR-DenseNet and the triplet networks in the multitask learning framework optimized by (15) is denoted as the SAR-Tri-DenseNet. To validate the efficiency and complexity of these networks, the total number of parameters for each model is summarized in Table IX. One can find that the proposed model has a moderate number of parameters.

For training, due to the large amount of parameters of the CNN-A, CNN-B, and VGGNet-16 and much deeper layers of the GoogLeNet, they are trained by the Adam optimizer [95], and the rest models are trained by the SGD with a momentum of 0.9. The initial learning rates of the CNNs A-D, VGGNet-16, and GoogLeNet are all set to 0.001 and the rest to 0.01. We set the total training epochs of the CNNs A-D as 300 and those of the AlexNet, VGGNet-16, GoogLeNet, and ResNet-18 as 90, 150, 90, and 90, respectively. The learning rate is divided by 10 at the epoch of 150, 200, and 250 for the CNNs A-D and at the one third and two thirds of the total epochs for the classical CNNs, respectively. All the comparing CNNs take the normal mini-batches of size 100 as inputs and are regularized with a weight decay parameter of 0.0005. Furthermore, the quantitative results of the 3- and 5-class classification tasks using different models are provided in Tables X and XI, respectively, and more analyses are as follows.

1) 3-Class Recognition Task: To inspect the experimental results using different network architectures in Table X, it is clear that the proposed methods and most of the comparing methods present promising results which can achieve an accuracy of conformably larger than 80% for the 3-class merchant vessels. It should be noted that the large amount of parameters in the very deep VGGNet-16 [28] leads to the lowest performance. Particularly, the shallower CNN-B [37], even with the maximum number of parameters, could acquire the best recognition rate of 83.30% among the state-of-the-art CNNs and ResNet-18 [30], with a more advanced structure, achieves highest accuracy of 83.44% in the classical CNNs. Furthermore, the proposed SAR-DenseNet network can obtain better recognition rate of 87.73%, which indicates the effectiveness of the densely connected architecture for CNN configurations to extract more discriminative deep representations. When integrating the triplet similarity constraint and the global Fisher regularization into the triple SAR-DenseNet architecture, the learned deep features have a larger interclass variance and smaller intraclass scatter. So the recognition rate using the SAR-Tri-DenseNet method can reach up to 88.97%, with a notable improvement of 5.67 points over the CNN-B.

For further evaluation and space limitation, Fig. 8 shows the confusion matrices of the best two comparing CNNs, i.e., CNN-B and ResNet-18, and the proposed SAR-DenseNet and SAR-Tri-DenseNet, from which some conclusions can be drawn. First, the recognition accuracy for each ship type of the proposed methods is better than or competitive with those of the comparing methods, which indicates the powerful ability of the proposed methods to extract pretty discriminative representations for different ship types. Second, the plain CNN-B tends to classify the ship types with fewer samples, e.g., the container ship type which only consists of 354 samples in the training data, into the ship types with more samples. While for the advanced ResNet-18 and the proposed SAR-DenseNet and SAR-Tri-DenseNet, the classification rate of the container ship can achieve 75%, 82.5%, and 85%, respectively, which can outperform the CNN-B by a biggest margin of 15%.

In conclusion, the designed task-specific network based on DenseNets [40] has endowed the proposed ship classification framework with great capacity to learn more discriminative deep features by propagating the deeply supervised signals to each preceding layer. Moreover, the incorporated metric learning scheme can further push the learned embeddings from different ship types far apart and pull those from the same type closer to each other in the embedding space. Hence, the proposed multitask learning framework is well suitable for ship classification in MR SAR

2) 5-Class Recognition Task: Extensive experiments of recognizing the five ship types are conducted to further testify

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TABLE VIII

DETAILED ARCHITECTURES FOR THE COMPARING CNNS. SPECIFICALLY, THE CONVOLUTIONAL LAYERS ARE DENOTED AS "CONV. (NUMBER OF FEATURE MAPS) AT (FILTER SIZE)" WITH THE RELU ACTIVATION FUNCTION [70]. THE FC LAYERS ARE DENOTED AS "DENSE (NUMBER OF UNITS)" WITH THE SIGMOID ACTIVATION FUNCTION [70]. THE MAX POOLING LAYERS ARE DENOTED AS "MAX POOLING (POOLING SIZE)—S2" WITH S2 INDICATING THE STRIDE OF 2 [70]. IN THE OUTPUT LAYER, C REPRESENTS THE NUMBER OF SHIP TYPES

CNN-A [66]	CNN-B [37]	CNN-C [38]	CNN-D [36]
Input (64 × 64)	Input (64 × 64)	Input (64 × 64)	Input (64 × 64)
Conv. 32@(2 × 2) - ReLU	Conv. 96@(3 × 3) - ReLU	Conv. 16@(5 × 5) - ReLU	Conv. 18@(9 × 9) - ReLU
Conv. 32@(2 × 2) - ReLU	Conv. 96@(3 × 3) - ReLU	Max Pooling (2 × 2) - s2	Max Pooling (6 × 6) - s2
Max Pooling (2 × 2) - s2	Max Pooling (2 × 2) - s2	Conv. 32@(5 × 5) - ReLU	Conv. 36@(5 × 5) - ReLU
Conv. 64@(2 × 2) - ReLU	Conv. 256@(3 × 3) - ReLU	Max Pooling (2 × 2) - s2	Max Pooling (4 × 4) - s2
Conv. 64@(2 × 2) - ReLU	Max Pooling (2 × 2) - s2	Conv. 64@(6 × 6) - ReLU	Conv. 120@(4 × 4) - ReLU
Max Pooling (2 × 2) - s2	Dense (1000) - Sigmoid	Max Pooling (2 × 2) - s2	Dense (120) - Sigmoid
Dense (256) - Sigmoid	Output (C) - Softmax	Conv. 128@(5 × 5) - ReLU	Output (C) - Softmax
Output (C) - Softmax	–	Conv. C@(4 × 4)	–
–	–	Output (C) - Softmax	–

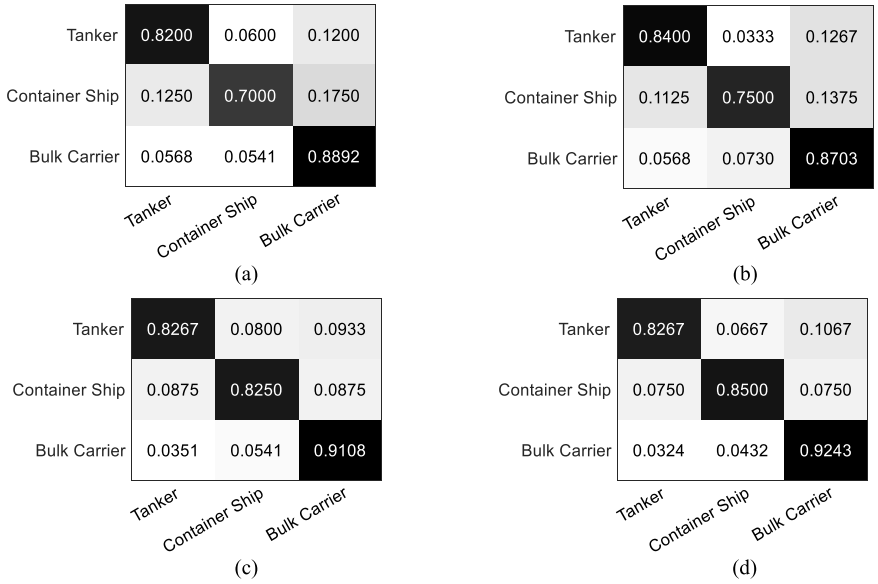


Fig. 8. Confusion matrices for the task of recognizing the tanker, the container ship, and the bulk carrier by using the methods of (a) CNN-B, (b) ResNet-18, (c) SAR-DenseNet, and (d) SAR-Tri-DenseNet, respectively.

TABLE IX

NUMBER OF PARAMETERS (M) FOR DIFFERENT NETWORK ARCHITECTURES

Model	Parameters	Model	Parameters
SAR-DenseNet	2.26	CNN-A	4.22
AlexNet	7.03	CNN-B	50.49
VGGNet-16	18.91	CNN-C	0.30
GoogLeNet	6.90	CNN-D	0.79
ResNet-18	11.06		

TABLE X

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 3-CLASS TEST DATA SET USING DIFFERENT NETWORK ARCHITECTURES

Network	Accuracy	Network	Accuracy
CNN-A	82.43 ± 1.61	AlexNet	81.63 ± 1.47
CNN-B	83.30 ± 1.27	VGGNet-16	72.37 ± 1.67
CNN-C	80.27 ± 1.35	GoogLeNet	83.30 ± 1.15
CNN-D	82.93 ± 1.57	ResNet-18	83.44 ± 0.90
SAR-DenseNet	**87.73 ± 0.42**	SAR-Tri-DenseNet	**88.97 ± 0.72**

the robustness of the proposed method. The overall accuracies and confusion matrices using different methods are shown in Table XI and Fig. 9, respectively. One can note that the overall accuracies of recognizing the five ship types have decreased dramatically compared to the results of the three ship types. This is mainly because the high similarity between different ship types and it is kind of challenging to extract effective representations of more ship categories under the limited resolution. For example, as shown in Figs. 6 and 9, the ship types of tanker, bulk carrier, and general cargo are

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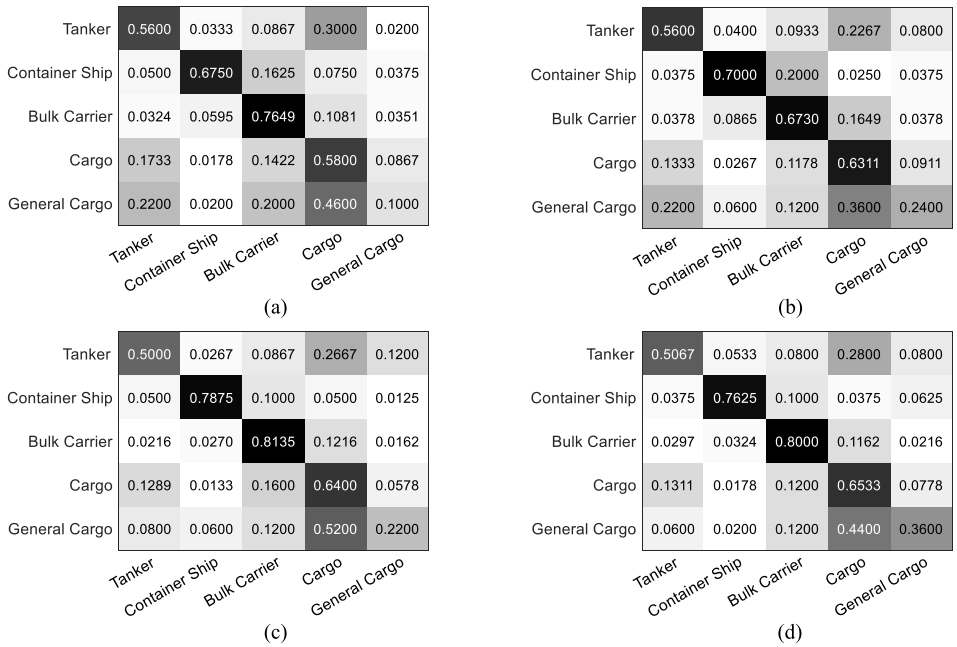


Fig. 9. Confusion matrices for the task of recognizing the tanker, the container ship, the bulk carrier, the cargo, and the general cargo by using the methods of (a) CNN-B, (b) ResNet-18, (c) SAR-DenseNet, and (d) SAR-Tri-DenseNet, respectively.

TABLE XI
OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE 5-CLASS TEST DATA SET USING DIFFERENT NETWORK ARCHITECTURES

Network	Accuracy	Network	Accuracy
CNN-A	60.96 ± 1.46	AlexNet	60.71 ± 1.40
CNN-B	62.51 ± 1.48	VGGNet-16	49.53 ± 1.80
CNN-C	60.20 ± 1.44	GoogLeNet	62.15 ± 1.61
CNN-D	60.13 ± 1.40	ResNet-18	63.44 ± 0.90
SAR-DenseNet	$\mathbf{66.71 \pm 0.46}$	SAR-Tri-DenseNet	$\mathbf{68.16 \pm 0.84}$

very similar to the cargo ships, thus many confusions exist. However, some valuable conclusions can also be drawn as follows.

First, considering the overall accuracies and the recognition rate of each ship type, the proposed methods could achieve better or competitive results over the compared plain and advanced CNNs. Second, the ship types of container ship, bulk carrier, and cargo ship are easier to distinguish due to the prominent appearance difference of the container ships and the large amount of the bulk carriers and cargos in the training data. Third, for the general cargo ships with least samples, the proposed methods and the ResNet-18 have better recognition performance than the plain CNNs, which indicates the superior capability of the advanced CNNs to recognize more ship categories. Noting that the general cargo ships are more apt to be classified as the cargos than the other ship types, which is rational in practice. Specifically, the total rates of the general cargo being classified as the cargo and general cargo types are 56%, 60%, 74%, and 80% by the CNN-B, ResNet-18, SAR-DenseNet, and SAR-Tri-DenseNet, respectively. Hence, the proposed methods are more effective to group the ship types with very similar appearance.

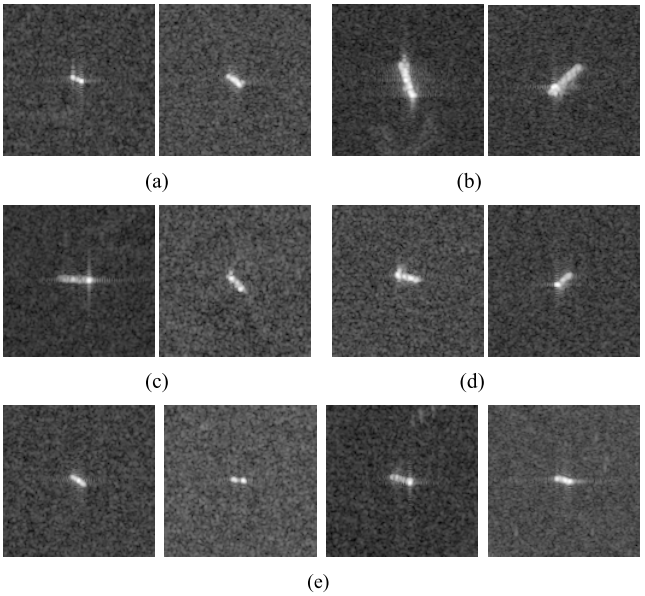


Fig. 10. Illustration of the classification results using the SAR-Tri-DenseNet. The first images of (a)–(e) are the correctly classified examples of the five ship types. The second images of (a)–(d) are the misclassified examples of the tanker, the container ship, and the bulk carrier; and the general cargo to the cargo, the bulk carrier, the cargo, and the cargo, respectively. The last three images of (e) are the misclassified examples of the cargo to the tanker, the bulk carrier, and the general cargo in sequence.

In addition, to intuitively interpret the results, some classification examples using the proposed SAR-Tri-DenseNet are shown in Fig. 10. From the confusion matrix in Fig. 9(d), one can find that the tanker, the bulk carrier, and the general cargo are mostly misclassified to the cargo ships, the container ship is mostly misclassified to the bulk carrier, and the cargo ships are generally misclassified to the tanker, the bulk carrier, and the

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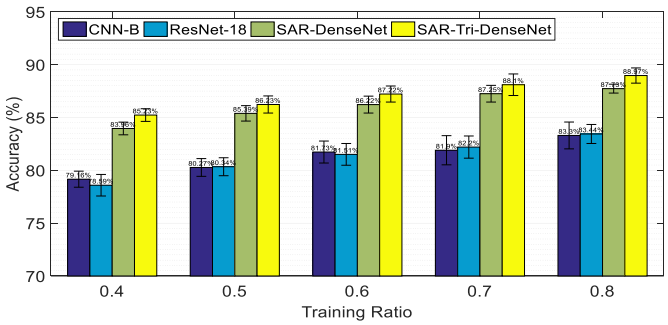


Fig. 11. Classification performance of different models with varying ratios of the training data to the whole data set.

general cargo, respectively. So, the similarities and differences among these ship types can be visually explored in Fig. 10.

G. Performance Evaluation Over Training Data Size

Generally, the superior performance of CNNs needs the guarantee of sufficient training data and a small number of training data may lead to overfitting. Thus, we conduct another experiment to test the robustness of the four representative methods abovementioned over the 3-class recognition task with varying training data size. The ratios of the training data to the whole data set are selected from the set of {0.4, 0.5, 0.6, 0.7, 0.8}, and the rest samples are used for testing. The preprocessing and experimental settings keep the same as aforementioned.

The results are intuitively illustrated in Fig. 11. One can find that the classification performance of the four models all degrade to some extent with the decreasing of the training data. The amounts of the accuracy degradation between the most and least training data by the CNN-B, ResNet-18, SAR-DenseNet, and SAR-Tri-DenseNet are 4.14%, 4.85%, 3.77%, and 3.74%, respectively, which indicates the proposed methods are more robust than the comparing methods over the varying training data size. Excitingly, the proposed multitask learning scheme SAR-Tri-DenseNet can achieve an overall accuracy of higher than 85% even with 40% of the data set for training (where the preaugmented numbers of the tanker, container ship, and bulk carrier are 270, 159, and 629, respectively). In summary, the proposed methods are more effective and have lower generalization errors to be applied for ship classification in MR SAR images.

H. Performance Evaluation Over Resolution Variance

In practice, it is important to ensure the robustness of the ship classification models to the resolution variance. This part will evaluate the performance of the four representative comparing methods on the SAR ship samples with different resolutions for 3-class recognition task. The multiple resolution versions of a ship image in the initial training and test sets are obtained by increasing the number of looks while keeping the pixel spacing fixed to the original 10 m, as similar to that in [70]. The initial resolution of the GRD images is about 20 m in both directions [18], and the ship samples with different reduced resolutions of about 40, 60, and 80 m are approximately simulated, as shown in Fig. 12. It is clear that the details of the ships are blurred with the decreasing of the resolution.

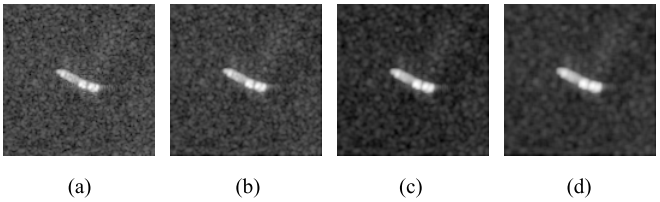


Fig. 12. SAR ship images at different resolutions with the fixed pixel spacing. (a) 20 m. (b) 40 m. (c) 60 m. (d) 80 m.

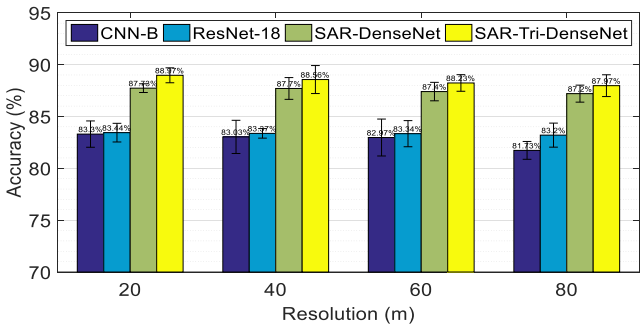


Fig. 13. Classification performance of different models with varying resolutions.

The training sets at different resolutions are preprocessed and augmented as aforementioned. We train the comparing methods at each resolution and evaluate them by the test data with the same resolution. The classification results are shown in Fig. 13, in which the evaluated methods have slight decrease in accuracy indicating the robustness to resolution variance. The proposed SAR-Tri-DenseNet has the best performance, and even at the lowest resolution simulated, the classification accuracy can still approach to 87.97%. It should be mentioned that the superior performance of the CNNs is guaranteed by the fully representative capability of the training set for the test conditions. In the future, more real multiple resolution SAR ship data sets should be collected and the performance difference between the GRD and the single look complex (SLC) SAR images for ship classification remains to be further explored.

VII. Conclusion and Discussion

This article proposes a multitask learning framework for ship classification in MR SAR images. To explore more effective feature extractors, a task-specific designed backbone network is developed inspired by the DenseNets. Our experiments prove that the proposed network is powerful to extract discriminative representations for effective MR SAR ship classification. To further boost the recognition performance, the triplet similarity constraint is combined with the softmax classification error penalty forming the multitask learning prototype, which can achieve good classification performance by pulling the deep representations coming from the same class closer to each other and pushing those of different classes far apart in the learned embedding space. To improve the generalization performance of triplet CNNs in the DML, the Fisher

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regularization term is imposed on the deep embeddings to take full advantage of the triplets in a training batch. Hence, the global information of the pairwise distances of the deep embeddings is fully mined and more robust models learned are obtained. Experiments on the 3- and 5-class recognition tasks demonstrate that the proposed method has better and more robust performance compared with the state-of-the-art and some advanced CNN models for ship classification. It is worthwhile to note that, for the same task of recognizing the three ship categories in the MR RADARSAT-2 SAR images using the naive geometric features in [64], the classification accuracy is only 71%, and the proposed method using the MR Sentinel-1 SAR images could achieve a mean recognition accuracy of as high as 88.97%.

Additionally, through our experiments, some remarks should be made. First, referring to Fig. 6, some typical artifacts exist in the collected SAR data set, such as the sidelobes, the azimuth smearing [18], and the strong sea clutters. Hence, some processing procedures, such as the target focusing and target super-resolution mentioned in [70], could be implemented in order to improve the performance further. Second, since the Sentinel-1 SAR GRD products are in VH–VV polarizations, it is thus important to utilize the polarimetric property of the data for better results. Third, it is necessary to collect much larger volume of MR SAR data to validate the robustness and the generalization ability of the proposed method, which is feasible as indicated in [96]. These aforementioned issues will be left for the future research.

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